

1. Administrative & Study Identification

Full Study Title: A Clinical Study of Sacituzumab Tirumotecan (MK-2870) in Patients With Bladder Cancer (MK-2870-027) **Short Title/Acronym:** TroFuse-027 / MK-2870-027 **Protocol Number:** MK-2870-027 **NCT Number:** NCT06637423 **Sponsor Name:** Merck Sharp & Dohme LLC **Investigational Product:** Sacituzumab tirumotecan (MK-2870 / SKB264 / sac-TMT) **Phase of Development:** Phase 1/Phase 2 **Medical Monitor:** Merck Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective **Primary Objective:** To evaluate the safety, tolerability, and dose-limiting toxicities of intravesical Sacituzumab Tirumotecan to establish the recommended phase 2 dose. **Secondary Objectives:** To evaluate pharmacokinetics, complete response rate (proportion of participants with the absence of visible tumors at the 12-week assessment), and the duration of complete response per local assessment.

2.2 Background & Rationale Standard treatment for patients with intermediate-risk (IR) non-muscle-invasive bladder cancer (NMIBC) includes transurethral resection of bladder tumor (TURBT) with or without adjuvant intravesical chemotherapy. However, 30% to 50% of patients experience disease recurrence, leading to repeated endoscopic resections and intravesical therapies that contribute to patient morbidity and decreased quality of life. Trophoblast cell-surface antigen 2 (TROP2) is broadly expressed in bladder cancer, making Sacituzumab Tirumotecan (a TROP2-directed antibody-drug conjugate) a viable targeted therapeutic alternative.

3. Study Design & Methodology

3.1 Design Framework **Study Type:** Interventional **Control Type:** Single-arm (Single Group Assignment) **Blinding:** Open-label (None) **Randomization:** N/A (Nonrandomized dose-escalation followed by expansion)

3.2 Planned Enrollment & Duration **Target \$N\$:** Approximately 32 participants in the Phase 1 dose-escalation part **Number of Sites:** Multinational **Estimated Study Start:** Late 2024 **Estimated Primary Completion:** TBD

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Has recurrent, locally documented, histologically confirmed low-grade (Ta) Non-Muscle Invasive Bladder Cancer (NMIBC). 2. Must have a visible tumor by cystoscopy within 12 weeks prior to first dose, and have not yet undergone TURBT. 3. Has intermediate-risk NMIBC defined as ≥ 1 of the following: multiple tumors, ≥ 1 occurrence of low-grade NMIBC within 1 year, early recurrence (< 1 year), solitary tumor > 3 cm, or failure of prior intravesical treatment. 4. Eastern Cooperative Oncology Group (ECOG) performance status of 0 to 2. 5. Urine cytology negative for high-grade urothelial carcinoma.

4.2 Exclusion Criteria 1. Active inflammatory bowel disease requiring immunosuppressive medication or history of inflammatory bowel disease (e.g., Crohn's disease, ulcerative colitis). 2. Uncontrolled, significant cardiovascular disease or cerebrovascular disease. 3. Known additional malignancy that is progressing or required active treatment within the past 3 years. 4. History of noninfectious pneumonitis/interstitial lung disease that required steroids or current active pneumonitis/ILD.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Sequential dose escalation of sac-TMT evaluated across designated cohorts. **Route of Administration:** Intravesical administration. **Duration of Treatment:** Treatment extends up to the 12-week cystoscopic assessment, continuing based on response.

5.2 Concomitant & Prohibited Therapy Permitted: Rescue medications for potential adverse events like stomatitis/oral mucositis (antihistamines, H2 receptor antagonists, acetaminophen, dexamethasone, antiemetics, antidiarrheal agents, etc.). **Prohibited:** Systemic immunosuppressants or concurrent investigational therapies not specified in the protocol.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures	:---
:---	:---	Screening	Day -28 to 0 Informed Consent, Medical History, Cystoscopy confirming visible tumor, Urine Cytology
Baseline	Day 0	First Dose (Intravesical administration), Safety baseline	Interim Weeks 1 - 11 Safety monitoring for Dose-Limiting Toxicities (DLTs), PK sampling
Assessment	Week 12	Cystoscopy to determine Complete Response (absence of visible tumors), safety follow-up	End of Study TBD Follow-up for Duration of Complete Response (DoR)

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Incidence of Dose-Limiting Toxicities (DLTs) and adverse events to establish the Recommended Phase 2 Dose (RP2D). **Measurement Method:** Clinical safety assessments graded by CTCAE criteria.

7.2 Secondary & Exploratory Endpoints * Complete Response Rate (CRR), defined as the absence of visible tumors at the 12-week assessment. * Duration of Complete Response (DoR). * Pharmacokinetics (PK) of Sacituzumab Tirumotecan.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring from consent through standard follow-up windows. Specific focus on mucositis, gastrointestinal issues, and pneumonitis/ILD. **Serious Adverse Events (SAEs):** Reported within 24 hours of site awareness to the sponsor. **Data Monitoring Committee (DMC):** Internal safety review committee evaluates DLTs before cohort dose escalations.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Safety Set (all participants receiving ≥ 1 dose of study drug); Efficacy Evaluable Set for tumor response. **Sample Size Justification:** ~32 patients allocated for Phase 1 dose escalation, sufficient to evaluate safety profiles and establish the RP2D. **Handling of Missing Data:** Standard handling for early withdrawals; missing 12-week cystoscopy assessments typically considered non-responders.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Routine onsite and remote monitoring to ensure protocol compliance and safety reporting accuracy. **Audit Trail:** Validated eCRF systems with secure audit trails and data clarification form mechanisms.

11. Study Identifiers & Governance

Primary ID: MK-2870-027 **Registry Number:** NCT06637423 **Sponsor/Lead Organization:** Merck Sharp & Dohme LLC **Study Title:** A Clinical Study of Sacituzumab Tirumotecan (MK-2870) in Patients With Bladder Cancer (MK-2870-027) **Official Regulatory Title:** A Clinical Study of Sacituzumab Tirumotecan (MK-2870) in Patients With Bladder Cancer (MK-2870-027)

12. Clinical Framework (Bladder Cancer Specific)

Primary Condition: Non-Muscle Invasive Bladder Cancer (NMIBC)
Diagnosis Threshold: Recurrent low-grade (Ta) intermediate-risk NMIBC with a visible tumor on cystoscopy
Medical Methodology: Intravesical targeted ablative therapy
Optimization Target: Dose optimization and reduction of disease recurrence without requiring TURBT

13. Study Procedures & Methodology

Management Pathway: Intravesical instillation of study drug directly into the bladder as an ablative monotherapy
Education Protocol (HCP): Safe handling and preparation of antibody-drug conjugates for intravesical delivery
Education Protocol (Patient): Informed consent detailing dose-escalation risks and required cystoscopy procedures
Quality Audit Mechanism: Ongoing review of DLTs during the dose-escalation phase

14. Observation & Follow-Up Schedule

Total Duration: Follow-up extends until disease recurrence, unacceptable toxicity, or completion of duration of response tracking
Onsite Visit Cadence: Baseline, regular treatment cycles, and protocol-mandated 12-week cystoscopy
Remote Monitoring Frequency: Between treatment cycles for ongoing AE tracking
Checklist Review Interval: Prior to every dose escalation decision

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				